Highlights

A signal-to-drift ratio for quantifying winding-fault detectability and selecting condition monitoring across three wind-turbine generator types

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- A model-free, detector-free signal-to-drift ratio quantifies fault detectability
- Phasor residuals remove the faulted-phase dependence of negative-sequence ratios
- Signatures relocate between voltage, controller and field current by machine type
- A decision table selects the cheapest adequate sensor suite with bootstrap confidence
- Learned detectors transfer across generator types under healthy-referenced features